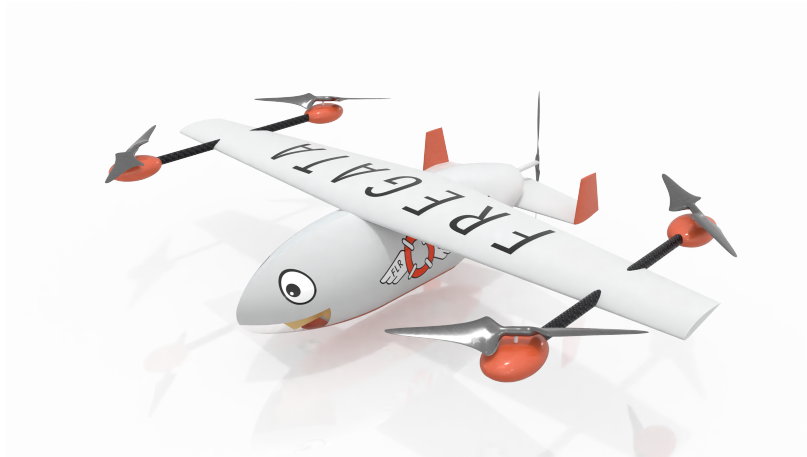




Flying Liferaft
Propeller Blade Aerodynamic Optimisation, Battery and ESC
Power Requirements



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Abstract

Over the past 5 years, there have been over 30,000 migrant deaths at sea. Nearly 15,000 of these occurred in the Southern Mediterranean migration route from Libya/ Tunisia to Europe. Many of these deaths are preventable if a rapid response UAV could provide temporary aid to these migrants to sustain them until surface vessels can reach them. This report presents the design of the propulsion and systems of a novel rapid-response UAV, designed to improve survival rates. It is intended that this report be read in conjunction with the rest of the Propulsion and System reports. After the initial sizing and baseline configuration were finalised, a decoupled parallel hybrid electric propulsion system was chosen. In particular, this report focuses on the aerodynamic design of the propeller blades using BEM theory, to achieve the mission profile defined in the executive summary[1]. Battery mass was a significant design constraint and therefore the majority of this report optimizes the VTOL propellers to reduce the required battery mass. The blades were successfully designed and validated using XROTOR, and significantly helped reduce the battery mass. Additionally, batteries and ESCs were selected to ensure sufficient power and energy can be provided to the propulsive systems.

FLR22

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1 Introduction

This report aims to outline the process taken to design the propulsion and systems configuration of a fixed wing VTOL UAV, designed to deliver a payload capable of helping people stranded at sea. The exact project definition can be seen in the executive summary, as well as the decision process taken in deciding the baseline configuration. The baseline configuration and mission profile is shown below for the ease of the reader.

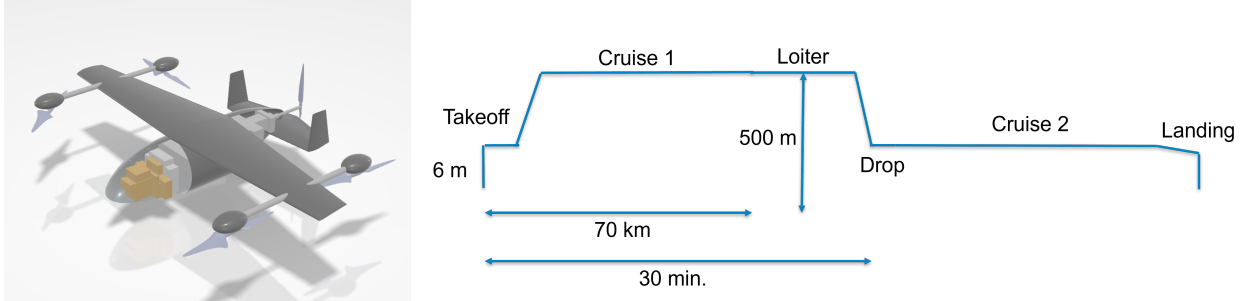


Figure 1: Baseline Configuration

The propulsion and systems team were directly responsible for designing a functional first iteration of the propulsion layout and design, in accordance with the initial sizing requirements. Additionally, the auxiliary systems and their layout within the aircraft were chosen and placed. Finally the weight and balance of the aircraft was determined. The workload was distributed among the group so that individuals had personal responsibility, however much of the design process was done in a collaborative manner. It is therefore imperative that this report be read in conjunction with the rest of the group's reports.

This report focuses on both the VTOL and pusher propeller aerodynamic optimization as well as helping to select appropriate motors, battery, engine and ESCs. This report is therefore split into two sections; propeller design followed by the associated systems.

2 Initial Sizing Propulsion Constraints

The first stage of the propulsion design process is to clearly define the thrust requirements of the aircraft. The constraint diagram (see [2]) of the baseline configuration is used as a basis for the design of the pusher propeller. As explained in the executive summary, these curves were obtained by defining the mission profile, and using empirical models to estimate the drag of the aircraft during its various stages of flight.

The constraint diagram produced in [2], shows that the most constraining flight condition is the climb segment as well as the stall speed. This used a climb rate of 500 ft min^{-1} and a speed of 58.3 ms^{-1} and demanded a minimum T/W of 0.22, or equivalently 111N given the MTOW of 51.5kg for the pusher propeller. The stall speed used was 30 ms^{-1} .

The thrust requirement for the VTOL propeller was estimated using a slightly different method. The primary function of the VTOL propellers is to provide sufficient thrust for takeoff and hover. The most demanding thrust requisite is for takeoff; a small vertical acceleration at MTOW. Additionally, during takeoff, the control surfaces are ineffective given the lack of forward velocity. Therefore, the controllability of the aircraft is also maintained by the VTOL propellers. Since it will be taking off from a coast guard vessel, the UAV must be operable in adverse weather conditions when the sea may be rough, resulting in the ship rocking significantly. It is therefore important that the UAV can quickly accelerate away from the ship deck to avoid be knocked by the ship in the initial stages of takeoff. Additionally, controllability at sea, where there could be significant gusts, requires a significant level of excess thrust to allow the propellers to maintain altitude and orientation. Therefore a T/W ratio of 1.2 was decided as sufficient to account for these requirements and was confirmed by the Performance, Stability and Control subteam. This meant each VTOL propeller was designed to provide at least 0.3T/W (151N per propeller).

3 Literature Review

The standard theory behind classic propeller design is blade element momentum theory (BEMT), and was introduced by Glauert[3] in 1927 (BEMT). In reality, the method capitalises on two independent formerly developed analyses completed in the 19th century by William Froude (1878) and William Rankine (1865). Froude first introduced the blade element method to calculate the forces on a turbine blade by discretizing the blade radially and then integrating the forces to calculate the total forces acting on the blade. Rankine developed a more generic concept to analyse the flow of a column of air through a wind turbine. Rankine expressed the problem as a simple conservation of mass and momentum exercise, and developed the fundamentals of actuator disk theory or axial momentum theory. Glauert combined these two analyses, as well as accounting for the influence of rotational effects, to create BEMT. 'The Blade Element Momentum method couples the momentum theory with the local events taking place at the actual blades'[4]. The derivation of the resulting equations is based on a light loading assumption whereby induced velocities are assumed small, and can be found in Design of Optimum Propellers [5]. A brief summary of the relevant equations is given below. These equations allow the thrust and torque forces to be calculated for a blade with a given geometry and set of operating conditions such as rpm and freestream velocity.

3.1 Governing Equations

The complexity in analysing a propeller blade is driven by the fact that the conditions change at each radial location along the blade. Since the blade is rotating, the tangential speed at each location changes and is calculated by equation 1.

$$v = \omega r \quad (1)$$

This means that the total velocity, freestream and rotational, seen by the radial station changes, as well as the effective angle of attack. This in turn means each blade element is exposed to a different dynamic pressure, as well as varying Reynolds numbers (depending on chord distribution too). These dependencies can be seen graphically in figure 2, as well as definitions of the important parameters ϕ , α , and β . The parameters a and a' are representative of the axial and tangential induction factor.

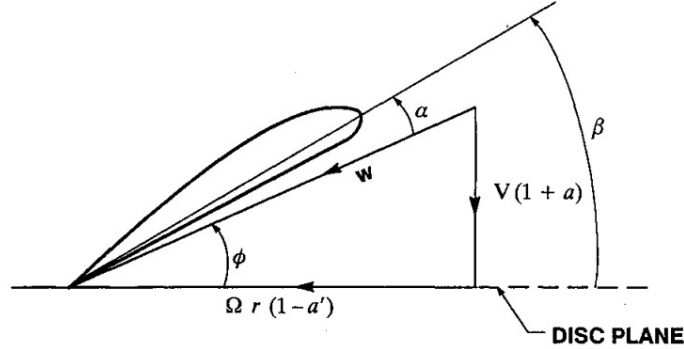


Figure 2: Blade station velocity triangles

With the geometry shown, the sectional thrust and torque is therefore given by equations 2 and 3, using Glauert's version of Froude's blade element method.

$$T' = L' \cos \phi - D' \sin \phi \quad (2)$$

$$Q'/r = L' \sin \phi + D' \cos \phi \quad (3)$$

Comparative equations using Rankine's axial momentum theory are given below.

$$T' = 2\pi r \rho V(1+a)(2VaF) \quad (4)$$

$$Q'/r = 2\pi r \rho V(1+a)(2\Omega r a' F) \quad (5)$$

In these equations the Prandtl tip loss factor is accounted for by the term F which is calculated using equation 6.

$$F = \left(\frac{2}{\pi}\right) \cos^{-1}(e^{-f}) \quad (6)$$

$$f = (B/2)(1 - \xi)/\sin(\phi_{tip}) \quad (7)$$

This term is essential and allows a simple model of the tip loss of lift caused by the high pressure air leaking around the tip of the blade to the suction side, in a similar fashion to tip loss on a standard 3D wing of an aircraft. These equations must be solved iteratively along the blade for a given twist and chord distribution.

3.2 Optimum Chord and Twist Distribution

Following from Glauert's work, Betz began working on not only analyzing propeller blades, but trying to optimise their shapes for a given set of operating conditions to minimize energy loss and hence power requirements to drive the shaft. Betz formally theorised that the optimum propeller geometry would be such that the inflow would be uniform, or equivalently, that the wake of the propeller acts as a helicoid. This can be mathematically shown to be equivalent to the condition that $r \tan \phi$ is constant at all radial stations (see [6] p.11). Therefore the BEM equations can be solved iteratively alongside the Betz condition to optimize the chord and twist distribution of a propeller blade for a given set of parameters.

4 Propeller Aerodynamic Optimization

After the initial sizing and baseline configuration, it was decided to adopt a hybrid electric propulsion system, whereby the VTOL propellers are powered by a battery, whereas the pusher propeller is powered by an internal combustion engine. Electric power supply to the VTOL propellers allow the RPM to be changed quicker to help improve controllability during takeoff, hover and landing. The details of this design decision can be found in [7]. Although reducing mass is always a priority in aerospace, it is especially important when dealing with electric propulsion systems, whereby battery mass contributes a significant fraction of the overall MTOW. Therefore, it is imperative that the VTOL blades are optimized for minimum power at their design point. Although less essential, the pusher propeller is also optimized to reduce fuel consumption, or equivalently, to increase range.

4.1 Outline Of Process

The optimisation of a propeller is notoriously difficult given the number of parameters that can be varied. Additionally, physical constraints of the specific design often restrict the optimisation. An example of this in this project was the requirement to keep the VTOL propellers less than a 0.3m radius so that the vehicle could fit in the 1.6m length constraint (including wing and two propellers). This constraint was generated by the need of the UAV to fit on a Tunisian coastguard vessel safely as discussed in the executive summary. Therefore the method that was taken was to develop a code to optimize the twist and chord distribution for a given set of parameters, and then the parameters were varied manually to evaluate the power requirements as well as balancing various design criteria that will be discussed. A completely analytic optimisation would be of little use due to the complexity of these varying and often contradictory design considerations.

4.2 Blade Geometric Optimisation

The generally accepted method to optimize the blade geometry for a given set of operating conditions, is to guess the axial and tangential induction factor, and iterate through the BEMT equations until the induction factors converge for the required thrust. A similar, equivalent method is set out by Adkins and Liebeck [5], instead iterating until ζ converges, where ζ is the displacement velocity ratio $\frac{v'}{V}$. This method is an adaptation of the Larrabee algorithm which can be seen in reference [6], and does not rely on the light loading assumption that Glauert's work initially relied on. This is essential especially for the VTOL propellers which will be operating at low axial velocities (vertical velocities), and therefore will have relatively large induced velocities relative to the freestream.

4.2.1 Adkins Algorithm

The Adkins algorithm assumes the following parameters are set before initiating the design: Blade diameter (B), hub radius (R_0), axial velocity (V), rotational velocity(ω), number of blades (B), thrust coefficient (T_c), and design lift distribution ($C_{l_{design}}$). For a proof of this algorithm see reference [5]. The following procedure is taken from the original work of Adkins and Liebeck [5], modified to include the equations.

- 1) Choose an initial estimate for ζ ($\zeta = 0$ will work).
- 2) Calculate values for ϕ and F at each radial station by Eqns. 6 and 8.

$$\phi = \tan^{-1}(\lambda/\xi(1 + \zeta/2)) \quad (8)$$

- 3) Determine the product Wc , (also helps calculate the Reynolds number) from the following equation.

$$G = \frac{Fr}{R\lambda} \cos\phi \sin\phi \quad (9)$$

$$Wc = \frac{4\pi\lambda G V R \zeta}{B C l} \quad (10)$$

- 4) Determine ϵ (which is $\frac{D}{L}$) and α_{design} from XFOIL airfoil data. These values are chosen to minimize ϵ .
- 5) Determine a , a' and W as below.

$$a = \zeta/2 \cos^2\phi (1 - \epsilon \tan\phi) \quad (11)$$

$$a' = \frac{\zeta R \lambda}{2r} \sin\phi \cos\phi \left(1 + \frac{\epsilon}{\tan\phi}\right) \quad (12)$$

$$W = \frac{V(1 + a)}{\sin\phi} \quad (13)$$

- 6) Compute the chord from step 3, and the blade twist $\beta = \alpha + \phi$.
- 7) Determine ζ , If this new value for ζ is not sufficiently close to the old one (e.g., within 0.1%) start over at step 2 using the new ζ .
- 8) Calculate the thrust and power and compare to design requirements to check for convergence.

Once this algorithm was implemented on MATLAB, the process of optimizing the other parameters began. The results from the code were validated against XROTOR (a software for rotor design and analysis developed by ESOTECH Developments[8]) thrust outputs for the same blade design to ensure the code was working properly. This can be seen in section 6.2, where the final aerodynamic blade (before structural modifications) was tested in XROTOR. Although XROTOR itself has an optimization algorithm which could have been used, XRotor does not accurately capture the true aerodynamic characteristics of the airfoil, rather relying on an assumptions of linearity from minimum lift to stall. Similarly, it implements an empirical model to evaluate the airfoil performance at different Reynolds numbers. The lift-drag model is similarly simplistic. The code that was implemented however, was designed to use actual XFOIL data at different Reynolds numbers and the true lift and drag curve for the airfoils used.

5 Propeller Design

The process taken to optimise the blade was highly iterative, and may be difficult to follow. Therefore an overview of the design process can be seen in figure 3. Each stage shown will be subsequently discussed in the report.

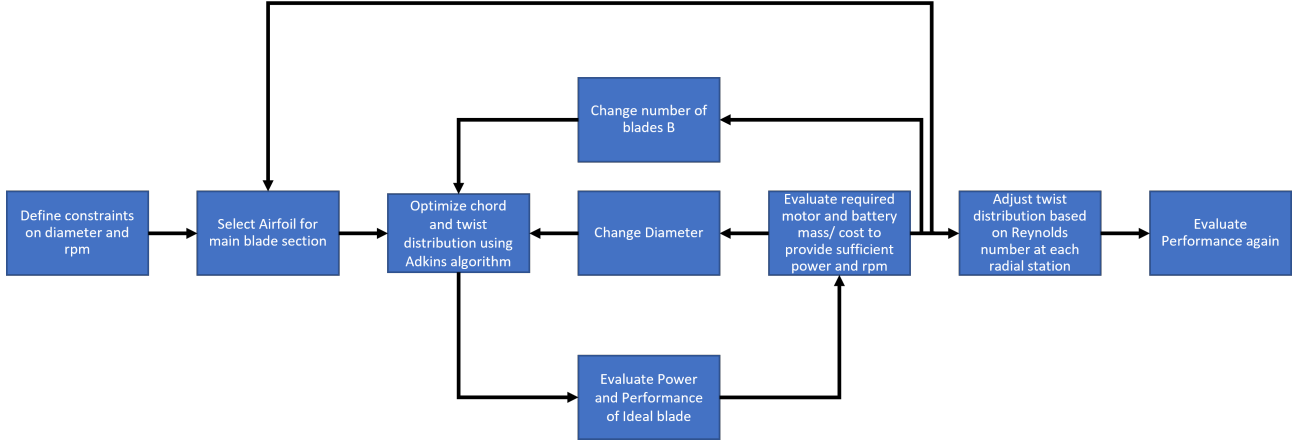


Figure 3: Method of Propeller Design

5.1 Constraints

The primary design requirement was that the pusher propeller could produce $0.22T/W = 111N$ and that the VTOL propellers could each produce $0.3T/W = 151N$. The free stream velocity this was optimised for was $58.3ms^{-1}$ (cruise velocity) and $2ms^{-1}$ (assumed vertical velocity during takeoff regime). Additionally, the radius of the VTOL propellers was restricted (halfway through the project; some of the iterations shown in future section still entertain the idea of a larger radius) to $0.3m$ as is discussed in section 5. Similarly the pusher propeller was designed so that it would not extend significantly beyond the bottom of the fuselage so that during landing and takeoff it would not hit the ship deck. When 'parked' on the deck the rotor blades would be locked in a horizontal position so that they will not hit the ground- this also restricted the pusher blade number to 2 blades, although the power implications were considered when making this decision and will be explained in section 5.4.

A somewhat more subtle and less intrusive constraint was on the rpm of the VTOL blades. Since the transition of the UAV from VTOL into cruise will occur with the aid of the VTOL propellers for lift until the aircraft reaches its stall speed, the blade tip speed may reach the M_{DD} , causing increased drag and some sonic effects. Therefore the rpm of the VTOL blade was restricted on all iterations so that the advancing blade tip in the transition period will not reach the predicted M_{dd} . Although it is not anticipated that the propeller will be on full throttle when the vehicle reaches stall speed, this was done as a precautionary measure in the worst case scenario. It could on the rare occasion reach this condition should a strong gust blow and require quick throttle commands to maintain control of aircraft.

Ultimately, the design objective was to minimise power and energy consumption, especially for VTOL, the battery for which contributed to 21% of the MTOW according to the initial sizing.

5.2 Airfoil Selection

The most important parameter of a suitable airfoil for a propeller is to have a high $\left(\frac{L}{D}\right)_{max}$. This is where the aerodynamic efficiency of the blade comes from and the main driver of the power requirement. Ideally, the twist distribution will allow the entire blade to be operating at the angle of attack that maximises the lift-to-drag. It is therefore also important that the lift coefficient at this $\frac{L}{D}$ is large. Another requisite is to ensure good airfoil characteristics at off-design operation, so that even when not operating at the design point, the blade will produce sufficient thrust for a reasonable amount of power. At off design operation, the rpm and freestream velocity will be different to the design point, and hence the local angles of attack seen by the blade at each station will differ from the angle it was designed to experience. Therefore it is important to ensure that the blade still has a good $\frac{L}{D}$, and crucially, does not stall significantly, which would result in a large loss of lift and thrust. Ideally the rpm to thrust model should be a smooth function allowing for stable and smooth controllability. Finally, the airfoil must have a thick $\frac{t}{c}$ ratio so that it has structural integrity, especially at the root. The following table 1 has tabulated some of the airfoils that were considered with their respective important characteristics.

Airfoil	$(\frac{t}{c})_{max}$	$C_{l_{max}}$	Angle/ °	$(\frac{t}{c})_{max}$	Angle/ degrees	Distance from stall	C_l at $(\frac{t}{c})_{max}$	$\min(\frac{t}{c})$ within 3 ° of $(\frac{t}{c})_{max}$	M_{DD}
VR12	0.106	1.21	11	39 (29 at low Re)	3	8	0.6	19	0.704
VR15	0.08	0.9	8	38 (28 at low Re)	3	5	0.4	10	0.75
NACA 0012	0.12	0.85	8	37	5	3	0.6	22	0.69
OA206	0.06	0.8	7	37 (28 at low Re)	3	4	0.4	20	0.77
OA212	0.12	1.2	10	43 (22 at low Re)	7	3	0.9	20	0.66
AH7476	0.059	1.6	10	75 (48 at low Re)	5	5	1.2	18	0.691
SG6043	0.1	1.55	10	66 (39 at low Re)	8	2	1.4	12	0.63
E216	0.1	1.5	10	70 (41 at low Re)	6	4	1.3	20+	0.67

Table 1: Airfoil Selection Data
[9]

The values for M_{DD} in the table were calculated using Korn's empirical model [10] shown in equation 14, where $Ka = 0.87$ for standard NACA type airfoils with low-medium camber and thickness (the type being used).

$$M_{dd} = Ka - \frac{t}{c} - 0.1C_L \quad (14)$$

The two standout airfoils that were chosen to test were the AH7476 and E216 airfoils for their excellent balance of characteristics. A comparison of these two airfoils for both the pusher and VTOL propellers is shown below in figure 4. Only one iteration is shown below for $R = 0.3m$ with $B = 2$ and varying rpm (this was the final iteration), however many similar comparisons were done for varying conditions and for other airfoils. The AH7476 originally was chosen for VTOL due to the reduced power consumption, but was later rejected due to its low $\frac{t}{c}$ ratio which resulted in the majority of the blade failing structurally. Therefore the E216 was chosen for both VTOL and pusher blades. In future iterations it would be worthwhile considering whether different airfoils may be used at the root/tip to optimize the balance between structural integrity and aerodynamic performance. Given the time constraints of this project, it was decided not to include this option at this stage.

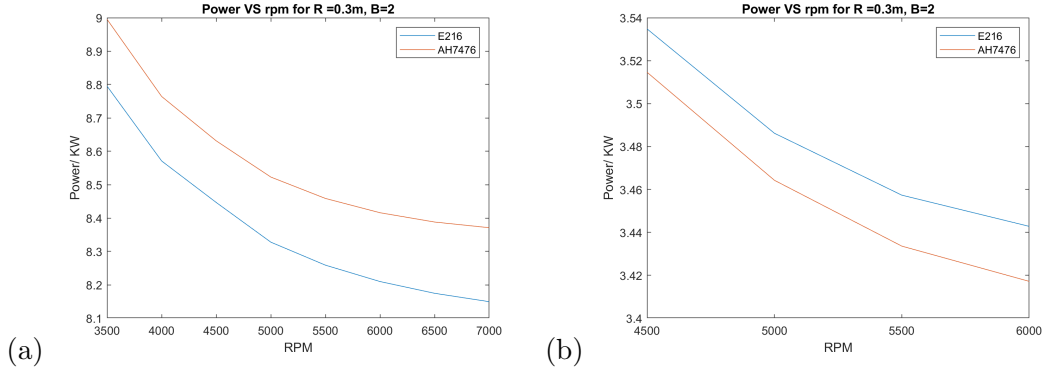


Figure 4: Airfoil Comparison
a) Pusher b) VTOL

5.3 Diameter of Rotor

The rotor diameter of a propeller has a large influence on the power requirements. Given the influence of tip loss, it is normally preferable to have as large of a diameter as possible to improve aerodynamic efficiency. Originally, no limit was set on the maximum diameter for either propeller.

Subsequently, the pusher propeller radius was qualitatively constrained to $0.3m$ so that it would not protrude beneath the fuselage extensively to avoid storage issues as well as safety when landing (even though it will be turned off completely and locked in horizontal position below a height of 15 meters, see also section 5.4). Even with this constraint, the increased power compared to $0.4m$ radius at the design rpm of 6500 is only $0.4KW$ (see figure 5), which is not significant considering the pusher is powered by a combustion engine. The reduced power would not enable a cheaper or lighter engine to be selected anyway (see reference [7] section 4.4), and would only reduce the fuel mass by 100g. Additionally, the ideal chord distributions shown in figure 5 further justifies this decision as the larger radii propeller designs had large region of the blade that would need to modification due to very small chord lengths which would cause structural and manufacturing issues.

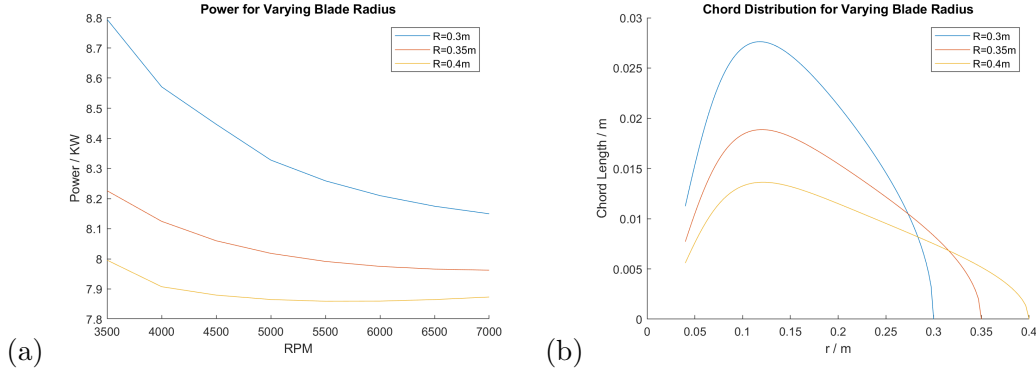


Figure 5: Influence of Radius on Power and Chord Distribution of Pusher Propeller (E216 Airfoil)

The main constraint on the VTOL propeller was actually manufacturing capabilities. Having a very long blade, would obviously not be manufacturable as the thickness of the chords for the required thrust will be too small. Additionally the moments at the root would be huge and therefore the blade will break. Therefore the following radii were evaluated as seen in figure 6. At this point in the project, the structural code had not been completed and therefore the AH7476 airfoil was still being considered as the main contender. A radius of $0.35m$ was chosen based on the compromise between power and manufacturability. Although the radius of $0.4m$ had a power saving of $0.9KW$, the ideal chord distribution meant that it was extremely thin (less than 2mm chord length from 20 cm along the blade onward). Increasing the radius to $0.35m$ increased the power by $0.3KW$ but maintained a chord of at least 25mm for the majority of the radius.

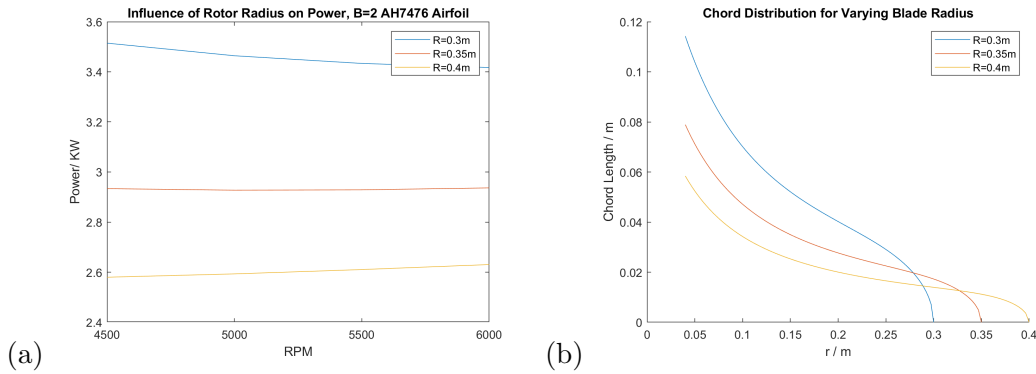


Figure 6: Influence of Radius on Power and Chord Distribution of VTOL Propeller

However, about halfway into the project, once a rough sizing of the wing had taken place, the propeller diameter for VTOL was constrained to $0.3m$ so that it could safely fit onto the Tunisian coast guard ship. Since the wing root was about $0.4m$ (see Aerodynamics team report), this meant that if the wing was to avoid blocking the propeller, the length of the UAV from front VTOL rotor tip to the back is $1.6m$, which would just fit onto the ship deck.

5.4 Number of Blades

Choosing the number of blades (B) for the two propeller designs was a compromise between power requirements, and geometric and aerodynamic constraints.

In particular, it was preferable for the VTOL to be a twin blade propeller, so that during cruise, when the VTOL propeller are idle, the blades would be forced by the drag into a streamlined position, reducing drag. In a three blade design, the blades would not all be in the streamwise direction and would therefore have significant extra drag and would experience higher loads. The power benefit of additional blades (as seen in figure 7) of only $0.13KW$ between 2 and 4 blades at the design RPM of 6000 did not justify these negative effects. This is especially true given the fact that the power requirement was the most constraining factor for the battery mass rather than

energy stored. Such a small decrease would not allow a lower mass battery to be used [11]. As mentioned before, the number of blades was varied iteratively for each airfoil and rpm; only the final design comparison is shown.

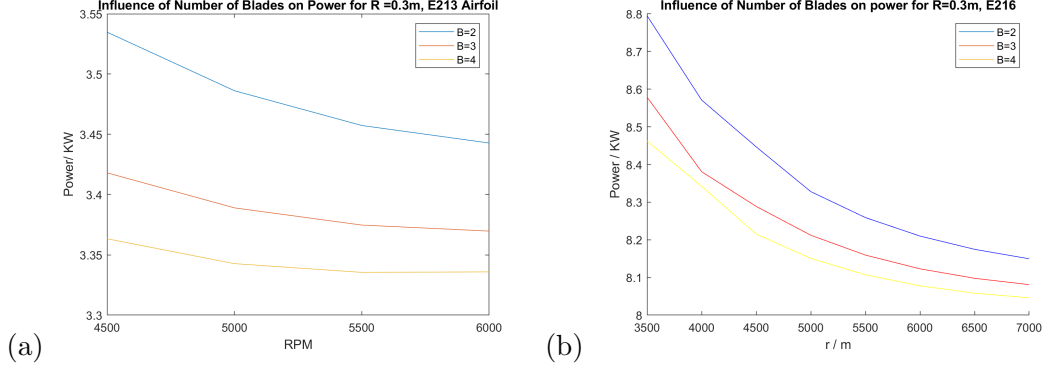


Figure 7: VTOL Number of Blades Optimization
a) VTOL b) Pusher

Additionally, the thrust model used (BEMT), does not account for rotor wake interference. Increasing the number of blades actually decreases the solidity as shown in figure 8, however it potentially increases the wake interference effects between the blades. Therefore, a 2 blade design was chosen given the minimal reduction in solidity relative to the interference more blades may present.

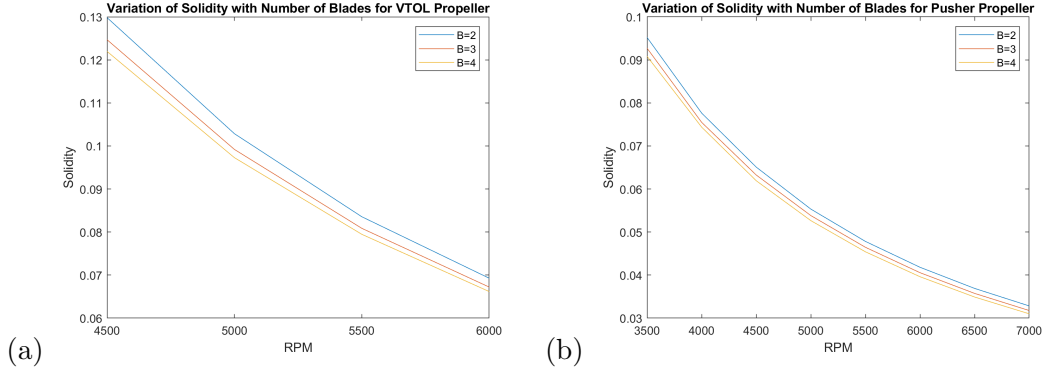


Figure 8: Influence of B on solidity σ
a) VTOL b) Pusher

5.5 Choosing an RPM

As can be seen in figures 5, 6, 7, and 8, increasing the rpm of the rotor reduces the power requirements. However, care must be taken to ensure the blade does not stall nor reach M_{DD} . This is particularly important for the VTOL propellers which will have an additional tangential velocity due to the aircraft's forward velocity during the transition. Therefore it was assumed that the VTOL propellers will be on full throttle until the aircraft reaches its stall velocity ($30ms^{-1}$) as a worst case scenario (may temporarily need full throttle for control reasons). Therefore, for a blade radius of $0.3m$ and the E216 airfoil (M_{DD} of 0.67), a maximum rpm of 6300 is allowed during forward flight at the stall speed. The pusher propeller is not subject to this combination of velocities and therefore the maximum rpm is defined so that the tip speed of the blade is below the M_{DD} of 0.67. This occurs at about 6850 rpm. Therefore a value of 6000 and 6500 was taken for the VTOL and pusher respectively, so that there is room for slightly extra throttle than the initial sizing had recommended. This is useful in adverse weather conditions.

It is difficult to predict where stall may occur in the initial design, and therefore was only checked after the final design was complete (see [2]).

Having discussed how the various parameters for each propeller were chosen, section 6.1 will discuss the final design and section 6.2 its validation.

6 Propeller ideal Performance

6.1 The Final Design

The final aerodynamic designs of the propeller blades before structural modification is shown below in figures 9 and 10. The main parameter choices are shown in table 2.

	Design Thrust / N	Number of Blades B	Airfoil	Cl.design	RPM	Diameter / m	Power Required / KW
Pusher	111	2	E216	1.37	6500	0.6	8.2
VTOL	151	2	E216	1.37	6000	0.6	3.4

Table 2: Main Propeller Parameters

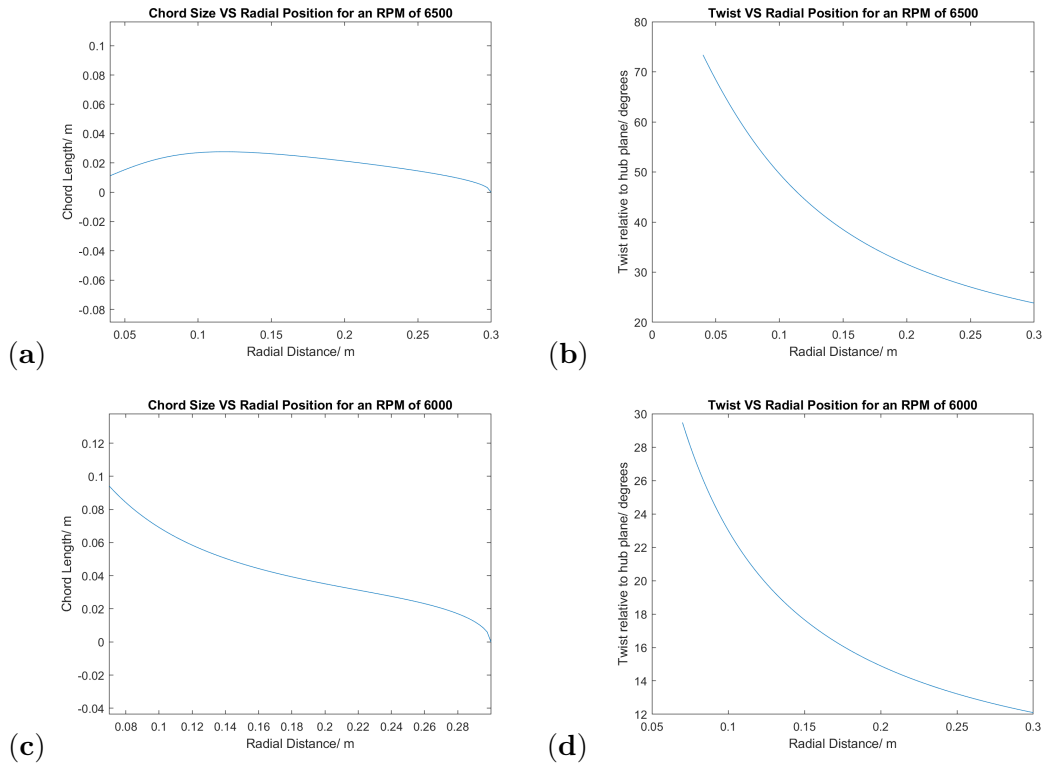


Figure 9: Final Aerodynamic Pusher Blade Distributions
a) Pusher Chord b) Pusher Twist c) VTOL Chord d) VTOL Twist

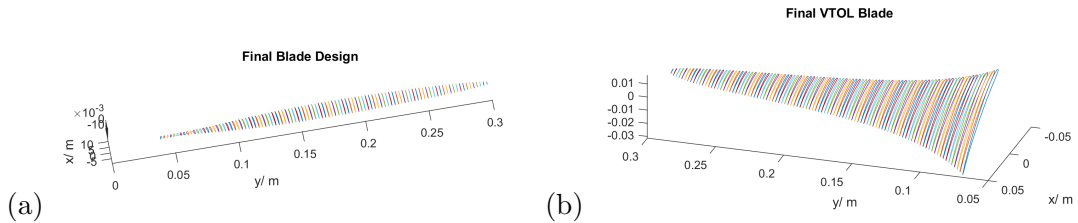


Figure 10: Final Aerodynamic Blade 3D View
a) Pusher Blade b) VTOL Blade

It was immediately clear from the chord distributions that both the pusher and VTOL needed modification. The pusher root is quite clearly too small, and will likely break structurally. Con-

versely, the pusher root chord is too large and therefore will need to be cut off near the root. With this in mind, on the last iteration of the design, an extra 5% thrust was added to both blades to account for the lost thrust post modification. Given that the majority of the thrust is generated away from the root of propellers, this was deemed sufficient. After the modifications this was confirmed[2].

Additionally, for the pusher blade, the sections nearest to the root will likely have reduced lift due to the presence of the fuselage in front of it. This will likely have two effects. Firstly, the presence of the fuselage distorts the streamtube airflow, and therefore reduces lift. Secondly, the propeller causes an increased velocity towards the back of the fuselage, as well as added rotation to the flow. This increases the dynamic pressure the fuselage encounters and hence increases the drag of the aircraft, reducing the effective thrust[12]. To account for these effects, two methods were used. Firstly, the blade was designed to have sufficient thrust from 0.07m to the tip, instead of from the hub of 0.03m (first 4cm of the blade was assumed to have no thrust). Additionally, a further 5% increase in thrust was designed for. It was further requested from the structures team to taper the aft fuselage as much as possible so that less of the pusher is blocked. It was suggested that the taper angle not exceed 20 – 25 degrees to avoid significant separation of flow from the fuselage which would further reduce thrust. The flow in this configuration should remain attached and laminar due to the favourable pressure gradient produced by the propeller itself.

It was later verified that separation was not significant even at high angles of attack in cruise by the Aerodynamics subteam (see [13]). The aerodynamics subteam completed a simulation using STAR CCM+ to obtain the flow characteristics around the fuselage and pusher propeller at various angles of attack. An image of the predicted flow field can be seen in the appendix figures 16 and 17. For more information regarding the simulation setup and assumptions, see reference [13]. It may be useful in future iterations of the design to explore boundary layer tripping devices mounted on the aft fuselage to try keep the flow attached to improve flow rate through the pusher propeller, or to reduce the aft fuselage taper.

6.2 XROTOR Validation

The main method used to ensure the blade was correctly designed to provide sufficient thrust was to compare results for the XROTOR software. This enables the blade geometry and airfoil characteristics to be inputted into a software that calculates the performance of the blade. The main comparison was the thrust, power and induced velocity values for both propellers when operating at their design points. This comparison can be seen in table 3.

		CODE	XROTOR	% Difference
Pusher at 6500 rpm	Thrust / N	111	125	+12
	Average Induced Velocity measured at rotor disk/ ms^{-1}	3.6	4	+11
	Power / KW	8.2	8.4	+2
VTOL at 6000 rpm	Thrust / N	151	143	-5
	Average Induced Velocity measured at rotor disk/ ms^{-1}	14.1	15	+6
	Power / KW	3.4	2.9	-15

Table 3: Validation of Code Against XROTOR

It must be noted that although there is some discrepancy between the two methods, this can be attributed to XROTOR’s rather simplistic model of an airfoils lift curve. It assumes a linear curve from zero lift angle of attack to stall, and uses an empirical model to estimate performance of the airfoil at different Reynolds numbers. The code however, extracted airfoil data from XFOIL at the correct Reynolds numbers, and used the actual lift curve. A comparison of these two curves can be seen in the appendix figures 15. Using these curves it also becomes apparent why XROTOR predicts higher thrust for the pusher propeller, but lower thrust for the VTOL one. This can be attributed to the lift distribution seen in appendix figure 11 and 13. It is clear that for the VTOL propeller, more of the lift is produced near the root and centre of the blade relative to the pusher propeller. The lift curves mentioned above show that XROTOR model predicts worse lift and drag characteristics at lower Reynolds numbers than the XFOIL model does and vice versa for higher Reynolds numbers. Therefore, less thrust is produced for the VTOL rotor on XROTOR than the code predicts. The opposite is true for the pusher propeller, where a more significant portion of the lift is toward the outer half of the blade; hence an over-prediction from XROTOR.

Additionally, the rpm required to produce sufficient thrust was also run on XROTOR; the results of which can be seen in the appendix figures 11 TO 14. The RPM required according to XFOIL was 6300 and 6166 for the pusher and VTOL respectively with a power of 8.4KW and 2.92KW, both close to the design point. These simulations further validate the method and results.

6.3 Final Aerodynamic Modification

A final modification was implemented on the blade to account for the airfoil's varying behaviour at different Reynolds numbers. The airfoil achieves its $\left(\frac{L}{D}\right)_{max}$ at slightly different angles of attack at different Reynolds numbers. Therefore the blade's twist was slightly modified to account for this and maintain the desired angle of attack. More details on this procedure can be found in reference [2].

7 Battery and ESC Selection

7.1 Battery and ICE Selection

The battery selection was an essential part of the UAV design process, given the significant weight it adds to the system. Although the VTOL propeller was not fully designed until late in the project, a rough estimate as to the battery requirements was required at an earlier stage. Therefore several assumptions were made following from the initial propeller design.

The aerodynamic optimization of the blade predicted a power requirement of 3.4KW for each VTOL propeller during VTOL operation (as opposed to hover which was 2.05KW). However after the Reynolds effects were included in the calculations (see [2]), the actual power requirements were reduced by 14.1% to 2.8KW. This was used to select the battery used to power all four motors with specifications shown in table 4. More details on the selection process can be found in reference [11].

Table 4: Main Battery Bank Configuration

Name of individual cell	18650-3030A	Battery Capacity (Ah)	27	Battery Voltage (V)	46.8
Number of cells in series	13	Total Weight (kg)	5.5	Current Discharge during VTOL (A)	239.3
Number of cells in parallel	9	Discharge temperature range (°C)	-20 to + 60	Current Discharge during hover (A)	173.5
Total number of cells	117	Dimensions of enclosure W x L x H (m)	0.24 x 0.16 x 0.065	Maximum discharge current (A)	270

Similarly, the internal combustion engine was chosen to provide sufficient power to the pusher propeller. After aerodynamic optimisation, the pusher required 8.2KW, but after Reynold's effects were included, the power was reduced by 4.8%, and only needed 7.8KW.

Table 5: Engine Selected

Engine	Engine Type	Mass (kg)	Power (kW)	SFC (g/kWh)
DLE120	Two-stroke piston	2.9	8.95	550

7.2 ESC Selection

Electronic speed controllers(ESC) act as in interface between control inputs and the motors[14]. It controls the RPM of the motor with the controller throttle command. With the battery configuration fixed, the current and voltage supplied to the motor driving the propeller at maximum rpm, is 242A and 46.8V. Often ESCs have two ratings for current, a current rating which is the maximum current it can sustain continuously, and a burst current; the current it can manage for short periods of a few seconds. Given the high current and voltage requirements of the motors, this is very important in the design.

Another important decision, is whether to use a 4-in-1 ESC, which is cheaper and reduces weight (this is minimal given the total mass of the drone). The problem with these ESCs, is that when one

of the 4 ESCs are damaged, all four must be replaced. Therefore, although it is cheaper on the first purchase to buy a 4-in-1, in the long run it is not necessarily the best option. Another negative of the 4-in-1 configuration relative to separate ESCs, is that having four individual ESCs decouples the motors entirely from each other, which is safer. Finally, given the high current requirements of the motors, overheating would be a significant issue in running the full 242A required for the four motors through one device- especially given the lack of heat dissipating airflow within the aircraft body. The fact that the UAV is designed to operate in the southern Mediterranean, exacerbates the heating of the ESCs, as temperatures can reach in excess of 35° Celsius in the summer months. Therefore, individuals ESCs were used- the specifications of which are seen in table 6.

ESC	ESC Configuration	Price Per unit/£	Total Weight/g	Max Continuous Current/A	Max Burst Current/A	Max Continuous Power/KW
KDE-UAS95HVC	Individual	180	456	95	165	4.3

Table 6: ESC Specifications

8 Summary

In conclusion, this report presents an overview of a full iteration of the design process of a rapid response UAV capable of providing aid to people in distress at sea. This report focuses on the design of the VTOL and pusher propeller for the aircraft. The report places specific emphasis on the aerodynamic optimisation to minimize energy consumption within the geometric and operational constraints of the UAV. Battery mass and fuel requirements were considered, as well as controllability of the aircraft. The Pusher and VTOL propellers have both been designed to provide 111N and 151N respectively. The maximum power required by the propellers is 2.8KW per VTOL propeller, and 7.8KW for the pusher. Additionally, appropriate batteries, combustion engine, and ESCs were selected to provide sufficient power for all operational uses of the propellers.

It is vital this report be read in conjunction with the associated reports ([2],[7],[11],[15]), as the final design of the UAV systems is based on the findings of all these reports. This report does not present a finished UAV system design, but rather a first iteration alone. In future iterations, more consideration should be placed on the influences of structural deformation of the blade, as well as further optimizations such as varying the airfoil along the blade. Additionally, the work presented is purely analytic/ computational, and should be verified in wind tunnel testing for actual performance.

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Appendix

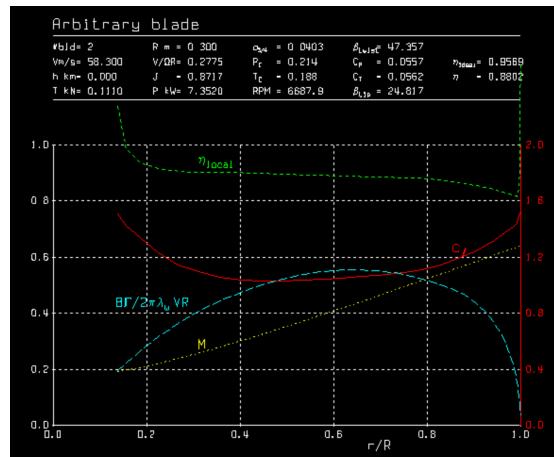


Figure 11: XROTOR Pusher Force Predictions

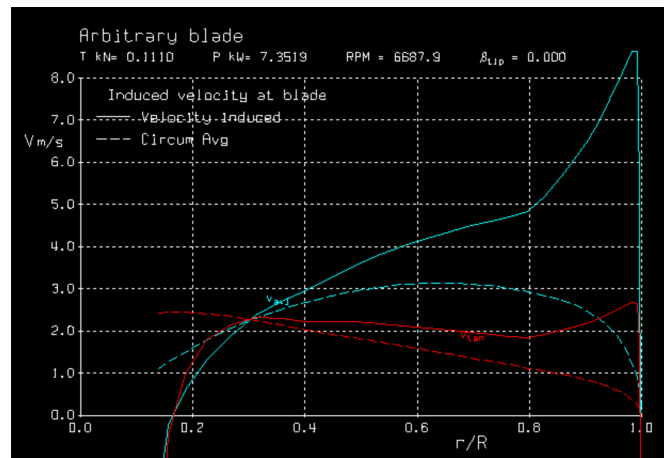


Figure 12: XROTOR Pusher Induced Velocities Predictions

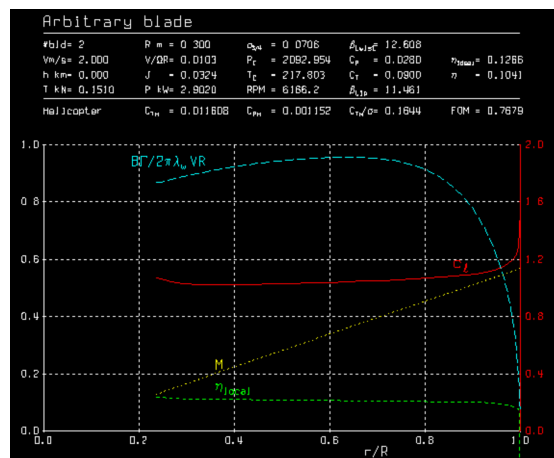


Figure 13: XROTOR VTOL Force Predictions

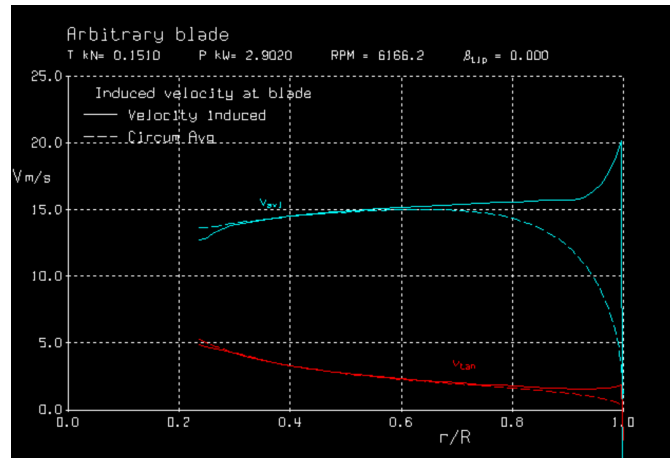


Figure 14: XROTOR VTOL Induced Velocities Predictions

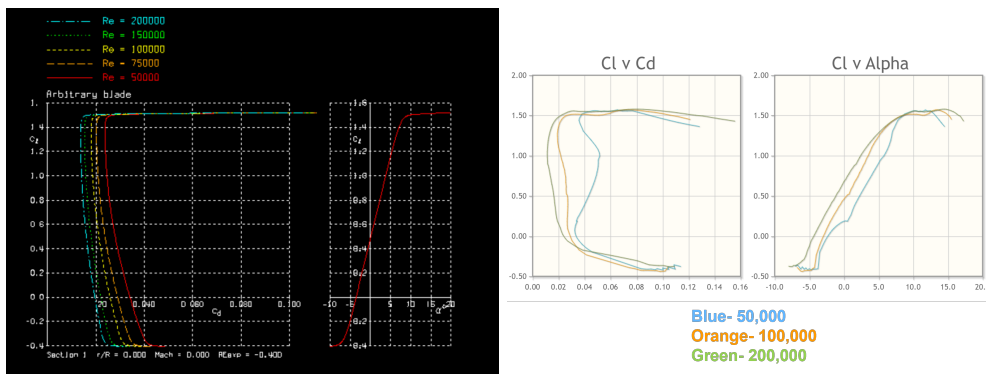


Figure 15: Lift Curve Comparison between XFOIL and XROTOR

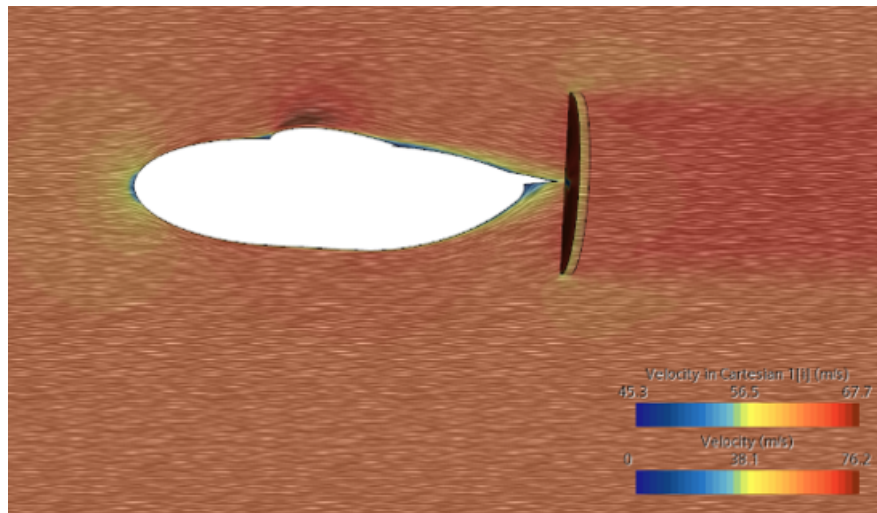


Figure 16: Flow field around fuselage at 3 degrees during cruise
 [13]

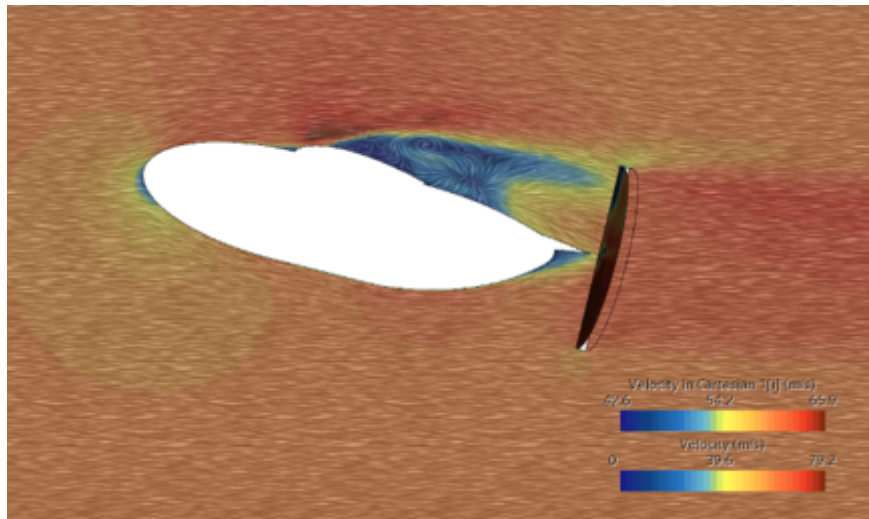


Figure 17: Flow field around fuselage at 14 degrees during cruise
[13]